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Effects of temperature, hypoxia, ammonia and nitrate on the bleaching among three coral species

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Abstract **Coral bleaching, which is defined as the loss of colour in corals due to the loss of their symbiotic algae (commonly called zooxanthellae) or pigments or both, is occurring globally at increasing rates, and its harm becomes more and more serious during these two decades. The significance of these bleaching events to the health of coral reef ecosystems is extreme, as bleached corals exhibited high mortality, reduced fecundity and productivity and increased susceptibility to diseases. This decreased coral fitness is easily to lead to reef degradation and ultimately to the breakdown of the coral reef ecosystems. Recently, the reasons leading to coral bleaching are thought to be as follows: too high or too low temperature, excess ultraviolet exposure, heavy metal pollution, cyanide poison and seasonal cycle. To date there has been little knowledge of whether mariculture can result in coral bleaching and which substance has the worst effect on corals. And no research was conducted on the effect of hypoxia on corals. To address these questions, effects of temperature, hypoxia, ammonia and nitrate on bleaching of three coral species were studied through examination of morphology and the measurement of the number of symbiotic algae of three coral species *Acropora nobilis*, *Palythoa* sp. and *Alveopora verrilliana*. Results showed that increase in temperature and decrease in dissolved oxygen could lead to increasing number of symbiotic algae and more serious bleaching. In addition, the concentration of 0.001 mmol/L ammonia or nitrate could increase significantly the expulsion of the symbiotic algae of the three coral species. Except for *Acropora nobilis*, the numbers of symbiotic algae of other two corals did not significantly increase with the increasing concentration of ammonia and nitrate. Furthermore, different hosts have different stress susceptibilities on coral bleaching.**

Keywords: ***Acropora nobilis*, *Palythoa sp.*, *Alveopora verrilliana*, temperature, hypoxia, ammonia and nitrate, coral bleaching.**

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Corals can be divided into reef-building corals and non-reef-building corals according to their ecological functions. The former contains symbiotic algae (commonly called zooxanthellae) but the latter not. In coral reefs, coral is made up of a lot of polyps, in which plenty of symbiotic algae inhabit. It is estimated that the population density of symbiotic algae is about one million cells per square centimeter in coral tissue. These symbiotic algae absorb carbon dioxide, phosphate and nitrate produced by corals and transform them into nutrition necessary to polyps. Therefore, it is absolutely necessary for corals to have their symbiotic algae[1—3].

Regular growth of corals needs some proper conditions, including suitable temperature, water transparency and adequate sunlight[4]. When corals are subjected to excess ultraviolet exposure, high temperature or other stress conditions, symbiotic algae will be expelled and corals will become white, which is called coral bleaching[5—8]. Coral bleaching is a signal of coral dying. If their living conditions cannot be improved in a short time, corals will die and expose the white calcium carbonate skeletons eventually. Coral bleaching, not only decolours corals due to loss of symbiotic algae and/or their photosynthetic pigments, making them more susceptible to disease, but also reduces biodiversity of coral reef ecosystems, even leading to the degradation of coral community and breakdown of coral reef ecosystems[5,9—15]. Therefore, it is the cornerstone to prevent corals from bleaching in protecting coral community and coral reef ecosystems.

Widespread coral bleaching events have been observed during the last two decades[6,11,16]. Coral bleaching can be induced by a variety of factors, including too low or too high temperature[17—21], excess ultraviolet exposure[18,22,23], high oxygen tension[10], heavy metal pollution[24], cyanide poison[25] and seasonal cycle[21,26,27].

Severe pollution occurs in the near shore waters in which reef-building corals inhabit with the rapid development of aquaculture, causing eutrophication in the seawater. Eutrophication can cause an increase in concentration of inorganic nitrogen and phosphorus, as well as blooms of bacteria and phytoplanktons, which can deplete the oxygen available to very low levels. To date, there has been little knowledge[26,28,29] of whether mariculture can result in coral bleaching and which substance has the worst effect on corals. And no research was performed on the effect of hypoxia on coral bleaching. To clarify these questions, effects of temperature, hypoxia, ammonia and nitrate on bleaching of three coral species were studied via examination of the morphology and the measurement of the amounts of symbiotic algae (commonly called zooxanthellae) of three coral species *Acropora nobilis*, *Palythoa* sp. and *Alveopora verrilliana*.

1 Materials and methods

(i) Collection and transient cultivation of corals.

This study was finished at Experimental Center of Tropical Biology, South China Sea, Institute of Oceanol-

ogy, the Chinese Academy of Sciences in October 6—19, 2003. Three coral species, *Acropora nobilis, Palythoa* sp. and *Alveopora verrilliana* were collected at Sanya Bay of Experimental Center of Tropical Biology, South China Sea, Institute of Oceanology of the Chinese Academy of Sciences, and cultivated transiently to ventilate in tanks in the room. Three coral species were cut into small similar-sized fragments and then rinsed clearly with 0.45 μm Millipore-filtered seawater for later use.

(ii) Effect of temperature on coral bleaching. Two temperature gradients, 25℃ and 32℃, were designed. Three different coral fragments were immersed in filtered seawater (0.45 μm) in equal volume in 500 mL beakers, respectively. The group at 25℃ were placed at room temperature (temperature range 25℃±2℃) and the other group were put in the water at constant temperature (32℃) under weak light. After stirring gently, a drop of solution were taken out and observed under microscope to measure the expulsion of symbiotic algae every 2 h.

(iii) Preparation for hypoxia solution[30] and its effect on coral bleaching. 0, 2.0, 4.0, 5.0 g sodium sulfite were dissolved in 1000 mL filtered seawater (0.45 μm) respectively and then added into a grain of potassium hydroxide. After shaking up, solutions in sealed reagent bottles were placed at room temperature for 2 h and different dissolved oxygen solutions were obtained (5 g/L sodium sulfite solution is anoxia). Three different coral fragments were immersed in the equal volume of the above-mentioned solutions in 500 mL beakers, respectively, and then placed at room temperature overnight.

(iv) Effect of ammonia and nitrate on coral bleaching. Five gradients of the concentration of ammonia and nitrate were designed, 0, 0.001, 0.01, 0.1 and 1.0 mmol/L respectively. 10.0 mol/L ammonium chloride solution and 10.0 mol/L potassium nitrate solution were made using filtered seawater (0.45 μm), then diluted into 0.001, 0.01, 0.1 and 1.0 mmol/L solution respectively. Three different coral fragments were immersed in the equal volume of the above-mentioned solution in 500 mL beakers, respectively, then placed at room temperature overnight.

(v) Determination of the content of dissolved oxygen. At the end of experiment, the content of dissolved oxygen of different dissolved oxygen solutions was determined by Quanta Hydrolab.

(vi) Measurement of the expulsion of symbiotic algae. After stirring gently, a drop of solution was taken out then placed on a slide. Under 10×10 amplification, eyeshots of well-distributed cell were selected to measure the number of symbiotic algae.

2 Results and discussion

(i) Coral bleaching and expulsion of their symbiotic algae. All the species of corals studied can release symbiotic algae. In the same environmental stress, the expulsion of symbiotic algae from different coral species was quite different, the largest amount is from coral *Acropora nobilis*, then from *Alveopora verrilliana* and the least from *Palythoa* sp. (Fig. 1). That indicated different hosts had different stress susceptibilities to expulsion of symbiotic algae[31]. Significant change of coral color occurred pre- and post-bleaching (Fig. 2).

Currently, it has been accepted that symbiotic algae from coral and anemone belong to a single pandemic species, *Symbiodinium* sp.[32—38]. However, they should be affiliated to different species[32,36], and Rudolf[32] also thought that different species of symbiotic dinoflagellates inhabited in different hosts. Symbiotic algae from anemone can be classified into zoochlorellae and zooxanthellae[35].

There were no apparent difference among the three species of symbiotic algae studied from different species of corals according to their morphologies and colors. Further research should be carried out to classify them clearly on the basis of their biological composition, karyotyping and phylogenies[36].

(ii) Effect of temperature on coral bleaching.

Generally, optimum growth temperature for corals ranges from 23℃ to 28℃ and expulsion of symbiotic algae occurs when the temperature of the seawater is out of this range. As shown in Fig. 3(a), few symbiotic algae were released from all the three coral species at room

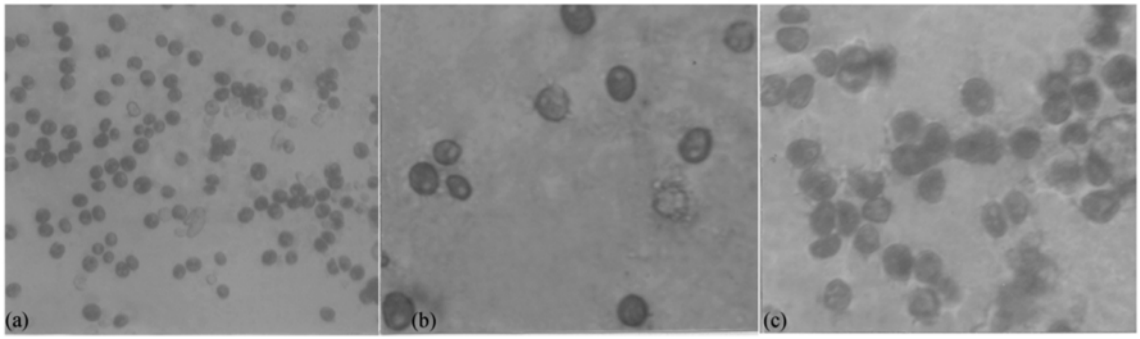

Fig. 1. Symbiotic algae of three coral species. (a) Acropora nobilis 10×10; (b) Palythoa sp. 10×40; (c) Alveopora verrilliana 10×40.

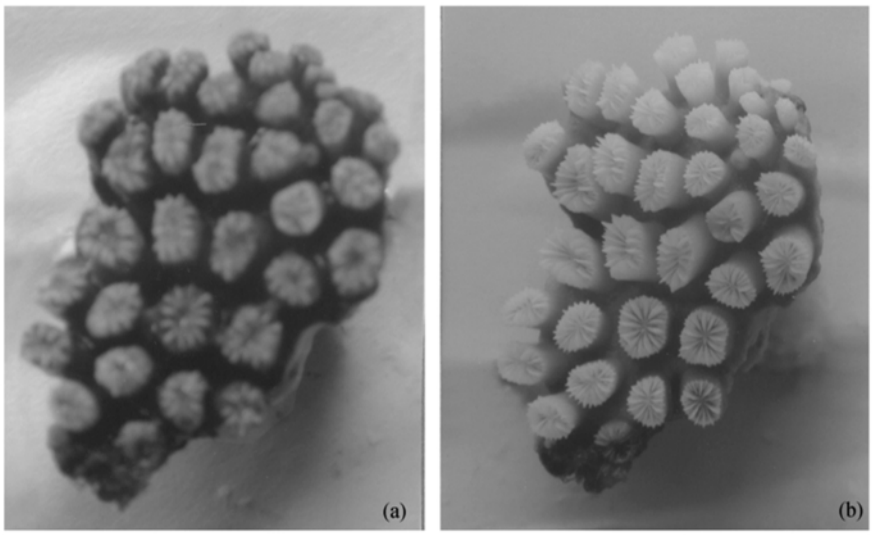

Fig. 2. Comparison of *Alveopora verrilliana* before and after bleaching by hypoxia solution.

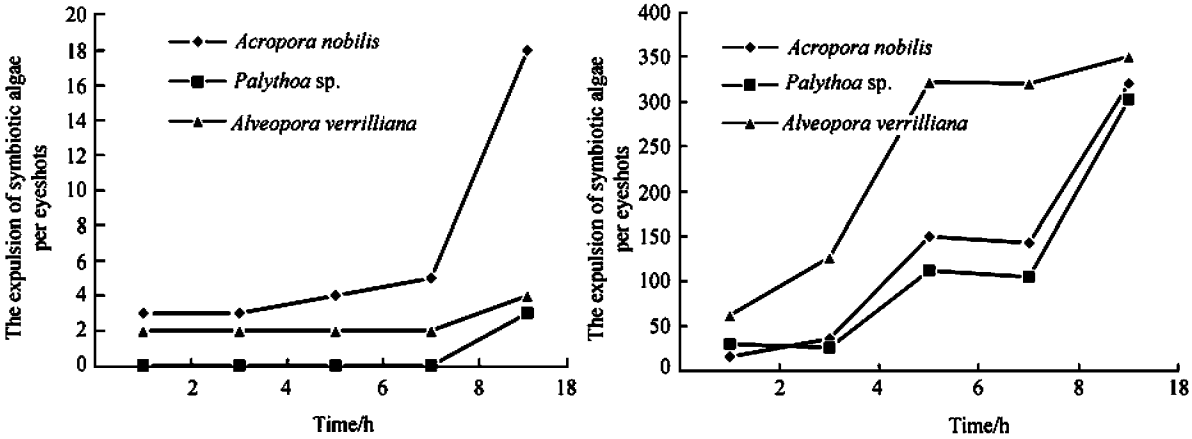

Fig. 3. (a) Effect of temperature on bleaching coral species (25℃); (b) effect of temperature on bleaching of three coral species (32℃).

temperature, but symbiotic algae of three coral species began to expulse gradually after 2 h at 32℃ and most of them were released after 18 h (Fig. 3(b)). Fig. 3(b) also shows that the coral *Alveopora verrilliana* is more susceptible to high temperature than the other two coral species and coral bleaching occurred in *Alveopora verrilliana* in the shortest time. Therefore, coral bleaching can be induced by higher temperature, which has been accepted by many scholars.

Photoinhibition occurs at higher temperature (generally above 30℃) in symbiotic algae in corals as a result of reduced efficiency in photosynthetic-electron transport[17,19,31]. That is to say, the photosynthesis of symbiotic algae could be impaired at higher temperature and it would be ceased completely at much higher temperature (above 34℃). Warner et al.[7] indicated that damage to photosystem Ⅱ in symbiotic algae was a determinant of coral bleaching at higher temperature. On the other hand, both superoxide dismutase (SOD) and catalase inactivate superoxide radicals (O_2^-) as well as hydrogen peroxide (H_2O_2) produced by symbionts (coral and symbiotic algae) at high temperature, which are mediators of oxygen toxicity, show an increase in being harmful to them with the accumulation of superoxide radicals (O_2^-) and hydrogen peroxide (H_2O_2)[39]. Therefore, it is an indicator for symbionts to subject to environmental stresses as expulsion of symbiotic algae occurs.

(iii) Effect of hypoxia on coral bleaching. Coral bleaching for three species significantly occurred in hypoxic solutions and the expulsion of symbiotic algae was increasing with the decreasing concentration of dissolved oxygen in solutions, suggesting that hypoxia played an important role in coral bleaching (Fig. 4). Fig. 4 also shows that *Palythoa* sp. was less sensitive to hypoxia than the other two coral species (*Acropora nobilis* and *Alveo-*

pora verrilliana) because the expulsion of its symbiotic algae was the least in the same concentration of dissolved oxygen solutions, which indicated that different hosts had different effect on coral bleaching[31].

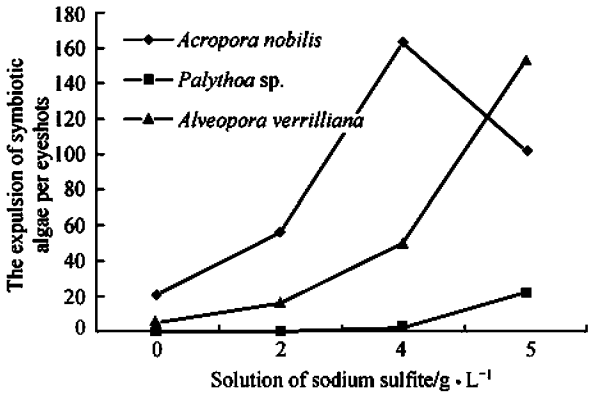

Fig. 4. Effect of different concentration of oxygen solution on bleaching of three coral species.

Sodium sulfite can decrease the solubility of oxygen in seawater[30]. In this experiment, addition of different quantities of sodium sulfite to seawater resulted in different hypoxic solutions. After symbiotic algae were quantified, the concentrations of dissolved oxygen in seawater were determined to be 2.95, 1.02, 0.11, 0.11 mg/L, respectively, resulting from additions of 0, 2.0, 4.0, 5.0 g sodium sulfite to seawater. Symbiotic algae will be expelled from corals if corals are smothered by shortage of oxygen for a long time. Moreover, symbiotic algae in corals require alkalescent environment (pH about 8.2) where their normal metabolisms are activated. However, corals will produce a lot of lactic acids by anaerobic respiration if hypoxia occurs in seawater, which cause acidic environment unfit for symbiotic algae to live.

The concentration of dissolved oxygen in seawater is normally 5 mg/L. However, in some coastal waters, several factors, such as eutrophication, sediments and particulate, and higher temperature, can deplete oxygen to very low levels and cause hypoxia which is defined as less than 2 mg/L dissolved oxygen in seawater. In Florida Keys, hypoxia and anoxia (no oxygen) have become more frequent[40,41] due to the increase of organic matters, which once led to mass coral bleaching in 1997. The stress of hypoxia can make corals more vulnerable to invasive organisms and corals will release symbiotic algae in order to reduce this danger. Under hypoxia, both photosynthesis and respiration significantly reduced in comparison with hyperoxia and normoxia[42]. In addition, superoxide dismutase (SOD) activity[43] reduces, and reoxygenation of hypoxic tissue could cause cellular severe damage[44]. As a result, the balance between symbiotic algae and hosts would be disrupted, resulting in coral bleaching.

(iv) Effect of ammonia and nitrate on coral bleaching. Nutrient enrichment in seawater has become more and more serious with the rapid development of mariculture in coastal waters. As shown in Figs. 5 and 6, 0.001 mmol/L concentration of ammonia or nitrate could significantly increase the expulsion of symbiotic algae. However, the expulsion of symbiotic algae of other two corals did not significantly increase with the increasing concentration of ammonia and nitrate compared to that of *Acropora nobilis*. Thus, different hosts have different stress susceptibilities to coral bleaching[31]. Figs. 5 and 6 also showed that the expulsion of symbiotic algae from both *Palythoa* sp. and *Alveopora verrillian* in the 0.001 mmol/L ammonia or nitrate was the most among the five different concentrations of ammonia or nitrate, which suggested that the growth of symbiotic algae in corals might be nitrogen-limited[28,29]. That is to say, the expulsion of symbiotic algae was decreasing rather than increasing when the concentration of inorganic nitrogen exceeded a certain value. Under nutrient enrichment, the

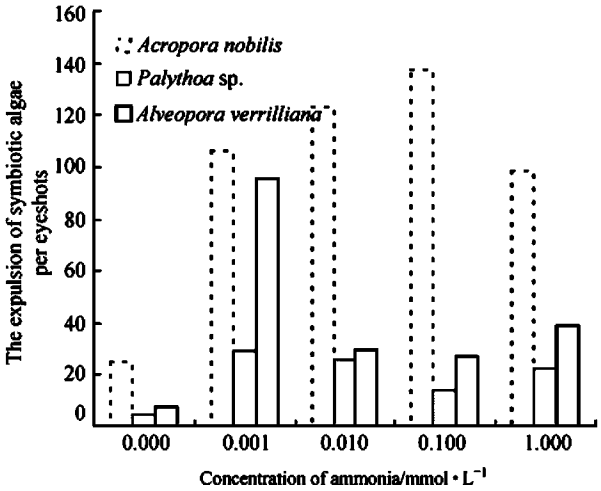

Fig. 5. Effect on ammonia on bleaching of three coral species.

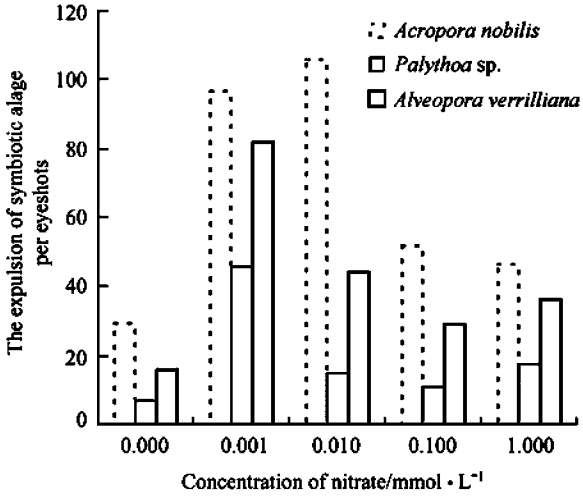

Fig. 6. Effect of nitrate on bleaching of three coral species.

host might be unable to control the growth of the symbionts, leading to the breakdown of the balance between the two compartments (hosts and symbiotic algae)[45]. However, regulation processes between the host and the symbionts are complex, since nutrient addition does not always induce an increase in the algal density[46,47].

Symbiotic algae make use of inorganic nitrogen to carry out photosynthesis. Muscatine et al.[28] thought that availability of inorganic nitrogen resulted in an increasing protein synthesis with the increasing the number of symbiotic algae, but symbiotic algae in corals can only consume limited nitrogen. Szmant et al.[29] also thought that some reef species conserved nitrogen by having relatively low rates of amino acid catabolism, so the growth of symbiotic algae within these corals might be inorganic nitrogen-limited. On the effect of inorganic nitrogen on coral bleaching, few studies were performed. Fagoonee et al.[26] reported that the expulsion of symbiotic algae was linearly proportional to elevated nitrate concentrations with a strong seasonal cycle, which is consistent with our study to some extent. All in all, symbiotic algae in corals may be inorganic nitrogen-limited, so it certainly results in coral bleaching with excessive inorganic nitrogen.

3 Conclusions

In the last two decades, coral bleaching has become more frequent and many scholars paid great attention to its causes. Little is known of whether mariculture can result in coral bleaching and which substance has the worst effect on corals. And no research was conducted on the effect of hypoxia on corals. In this paper, the effects of temperature, hypoxia, ammonia and nitrate on bleaching of three coral species were studied. The results show that hypoxia, ammonia and nitrogen all can result in coral bleaching.

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